

LLMs as Economic Measurement Tools

A Framework for Job Posting Classification

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What Researchers Really Need

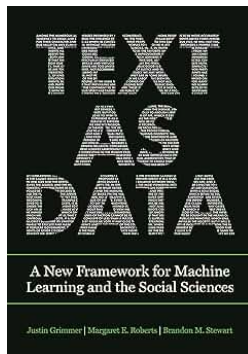


Figure: Text as Data

- The ability to measure interesting concepts freely and efficiently.
- Allows us to start from research idea to data instead of reverse.

Motivation: Job Posting Data

Two complementary research streams:

Stream 1: Multidimensional Skills (Deming and Kahn, 2018; Atalay et al., 2020; Hassan et al., 2025)

- Beyond education & occupation
- What skills do firms require?

Stream 2: Workplace Flexibility (Adams et al., 2019; Hansen et al., 2023)

- Non-wage amenities matter
- Remote work arrangement rises
- What workplace conditions matter?

⇒ LLMs + Job postings could measure both dimensions at the firm-monthly level

Our Framework

1. Create a **taxonomy**: 25 skill and workplace flexibility categories
2. Design **Prompt Engineering**: Ask LLMs for binary classification + evidence for each category
3. **Benchmark Construction and Testing**: Compare LLM classifications to human annotations (986 postings) and keyword baselines
4. **Threshold Tuning**: Exploit log probability thresholds to mitigate false positive cases
5. **Finetuning**: Finetune GPT-4.1-mini using OpenAI API
 - **Validation 1 – Explanatory power**: Do skill measures obtained by LLMs predict firm productivity better?
 - **Validation 2 – Survey**: Do WFH measures correlate with survey results?

Step 1: Building the Taxonomy

- **Digital Skills (10 categories)** (Nania et al., 2019): e.g., Programming, Data Analysis, GenAI
- **General Skills (7 categories)** (Deming and Kahn, 2018): e.g., Social Skills, Cognitive Skills, Customer Service
- **Workplace Flexibility (8 categories)** (Adams et al., 2019): e.g., Remote Work, Flexible Hours, Part Time

Each posting receives 25 binary labels: 1 = skill/amenity present, 0 = absent

Step 2: LLM Prompting Strategy

Main Prompt: Binary Classification + Evidence

Input

- Job posting text
- Category definitions
- Requirement for JSON output

Output

- Binary decision (0/1)
- Supporting text evidence (enables intensity measures in future work)
- Token-level log probabilities

Other Prompt Strategy	Description
Binary Classification	Only output 0/1 predictions
Two-Shot Decomposition	Separate API calls for skills vs. flexibility

Step 3: Benchmark Construction and Testing

Ground Truth: 986 Human Annotations

- Native speakers manually coded postings
- Two annotators marked relevant text spans
- No resources for cross-coder analysis

Home / argilla / DEMO1

1 record selected

100% 1 - 10 of 22

Text

Senior Business Analyst to Financing Businesses A tribe that provides fast and relevant financing through simple processes to support our Business customers across the Nordic. In Financing Businesses Tribe, we are committed to be at the forefront of digital development in our pursuit to create the bank of the future. To do this we need **bright minds and highly motivated individuals with experience from the banking sector and courage to challenge the way we do things**. You will join Financing Businesses tribe, anchored in Copenhagen, Vilnius, and Bangalore, with the aim to significantly improve financing processes for business customers in the Nordics. The position is a unique opportunity to contribute to a non-traditional banking set-up, high pace, agile development, flat hierarchy, and with an overarching customer-focused approach til we have fun every day and most important we celebrate our successes. The position can be based in Denmark, Sweden, Norway, Finland, or Lithuania. In spring 2024, we will move our Danish location to a new domicile on Postgrunden next to Copenhagen Central Station. Solid banking knowledge and a desired to improve status quo You will join a team of 5-6 highly dedicated Business Analysts in the Financing Solutions squad, which are responsible for the credit and loans products, as well as the entire credit processes for Business customers across the Nordic. As a Business Analyst your

Classify the tokens according to the specified categories. *

Search labels

- 1 wfhOptions
- 2 flexibleHours
- 3 flexibleTasks
- 4 temporary
- 5 partTime
- 6 shiftWork
- 7 traveling
- 8 CRM
- 9 computerSupportAndNetworking
- 10 dataAnalysis
- 11 digitalDesign
- 12 digitalMarketing
- 13 machiningAndManufacturingTechnology
- 14 productivity
- 15 programmingAndSoftwareDevelopment
- 16 characterTraits
- 17 cognitiveSkills
- 18 customerService
- 19 financialSkills
- 20 managementSkills
- 21 socialSkills
- 22 writingAndLanguage

Discard Save as draft Submit

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Finding: Open-Source Parity

Model	Accuracy	Macro F1
<i>GPT Family</i>		
GPT-4.1-mini	85.3%	0.673
GPT-4.1	82.4%	0.662
GPT-4.1-nano	84.5%	0.627
<i>Open-Source</i>		
Llama 3.3 70B	85.3%	0.652
<i>Baseline</i>		
Keyword Methods	73.4%	0.425

- Open-source models match GPT Family performance
- LLMs beat keywords
- Model size matters

Finding: Prompt Engineering Has Limited Impact

Prompting Strategy	Accuracy	Macro F1
Binary Classification	86.0%	0.675
Two-Shot Approach	85.9%	0.675
Binary + Evidence	85.3%	0.671

All strategies evaluated using GPT-4.1-mini on 986 annotated postings

Step 4: Model's "Confidence" and Threshold Tuning

Headline: Tennis Champion Showcases Hidden Talents in Symphony Orchestra Debut

Output token 1: Art, logprobs: -0.028133942, linear probability: 97.23%

Output token 2: Sports, logprobs: -4.278134, linear probability: 1.39%

Figure: Log Probability Use Case from OpenAI

Applied to our case:

WFH: [?], the next token could be 0 or 1.

- **Output token 1:** 1 (confidence: 92%)
- **Output token 2:** 0 (confidence: 8%)

So we take Output token 1's confidence score as the confidence measure we use.

Finding: Confidence Thresholding Improves Accuracy

Idea: Reject low-confidence positive predictions.

Example:

In Job post 1, WFH: 1 (confidence: 92%) → Keep
In Job post 2, WFH: 1 (confidence: 65%), Threshold: 80% → Flip to WFH: 0

Model	Accuracy	Macro F1
GPT-4.1-mini (Baseline)	85.3%	0.673
Threshold Tuning Fold 3	91.2%	0.677
Threshold Tuning Fold 2	90.3%	0.639
Threshold Tuning Fold 1	90.3%	0.632

Three-fold cross-validation. Each fold uses 657 postings for threshold calibration and 329 for testing

Step 5: Finetuning via OpenAI API

Process:

1. Format 657 postings as training pairs (job text + human annotations) in JSONL
2. Call OpenAI finetuning API with GPT-4.1-mini
3. Wait 2 hours for training to complete
4. Evaluate finetuned model on held-out 329 postings
5. Rotate and repeat for each fold

Average accuracy: 93.0%, Macro F1: 0.712. Finetuning achieves highest performance.

Validation 1: “Explanatory Power” Validation

Steps

- 4.3 million job postings (2007–2023)
- Processed with GPT-4.1-mini
- Aggregated at firm level
- Merged to Danish firm registry using CVR number

Empirical Specification

$$\log(\text{Revenue per Employee}) = \beta_0 + \beta_1 \text{Skill Measures} + \beta_2 \text{Controls} + \theta_j + \epsilon$$

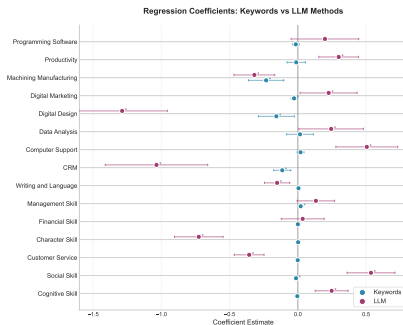
where θ_j denotes industry fixed effects

Findings for Validation 1: The Productivity Validation

Results: Explanatory Power

- Baseline (no skills): $R^2 = 0.470$
- + Keyword-based skills: $R^2 = 0.478$ (gain: 0.8%)
- + LLM-based skills: $R^2 = 0.497$ (gain: 2.7%)

LLM measures explain **3× more** productivity variation than keywords



Validation 2: Survey Validation

Labor Market Survey (AKU)

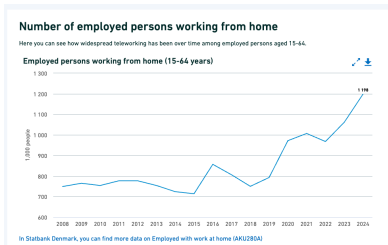


Figure: WFH measure from survey

LLM-Based Job Postings

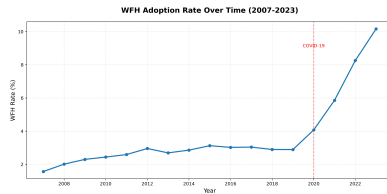


Figure: WFH measure from LLM classification

Independent validation: LLM measures track survey-based labor market trends

Key Contributions

Methodological Advances

- Open-source models (Llama-3.3-70B) match commercial APIs while enabling on-premise deployment
- Log probability thresholds improve accuracy to 91% without retraining
- Finetuning achieves 93% accuracy through three-fold cross-validation

LLMs transform labor market text into structured economic measures with accuracy and economic relevance.

- Adams, A., Jensen, M. F., and Petrongolo, B. (2019). The contribution of employee-led and employer-led work flexibility to the motherhood wage gap.
- Atalay, E., Phongthientham, P., Sotelo, S., and Tannenbaum, D. (2020). The evolution of work in the united states. *American Economic Journal: Applied Economics*, 12(2):1–34.
- Deming, D. and Kahn, L. B. (2018). Skill requirements across firms and labor markets: Evidence from job postings for professionals. *Journal of Labor Economics*, 36(S1):S337–S369.
- Hansen, S., Lambert, P. J., Bloom, N., Davis, S. J., Sadun, R., and Taska, B. (2023). Remote work across jobs, companies, and space. Technical report, National Bureau of Economic Research.
- Hassan, T. A., Hollander, S., Kalyani, A., van Lent, L., Schwedeler, M., and Tahoun, A. (2025). Text as data in economic analysis. *Journal of Economic Perspectives*, 39(3):193–220.
- Nania, J., Bonella, H., Restuccia, D., and Taska, B. (2019). No longer optional: Employer demand for digital skills.